



Real-time control using Bayesian optimization: A case study in airborne wind energy systems[☆]



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ABSTRACT

This paper presents a framework by which a data-driven optimization technique known as Bayesian Optimization can be used for real-time optimal control. In particular, Bayesian Optimization is applied to the real-time altitude optimization of an Airborne Wind Energy (AWE) system, for the purpose of maximizing net energy production. Determining the optimal operating altitude of an AWE system is challenging, as the wind speed constantly varies with both time and altitude. Furthermore, without expensive auxiliary equipment, the wind speed is only measurable at the AWE system's operating altitude. In this work, Gaussian Process modeling and Bayesian Optimization are used in real-time to optimize the AWE system's operating altitude efficiently, without the use of auxiliary wind profiling equipment. Specifically, the underlying objective function is modeled by a Gaussian Process (GP); then, Bayesian Optimization utilizes the predictive uncertainty information from the GP to determine the best subsequent operating altitude. In the AWE application, context-dependent Bayesian Optimization is used to handle the time-varying nature of the wind shear profile (wind speed vs. altitude). Using real wind data, our method is validated against three baseline approaches. Our simulation results show that the Bayesian Optimization method is successful in significantly increasing power production over these baselines.

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1. Introduction

Wind energy is one of the most promising renewable resources for displacing traditional fossil fuel sources. Airborne Wind Energy (AWE) systems are a new paradigm for wind turbines, in which the structural elements of conventional wind turbines are replaced with tethers and a lifting body (a kite, rigid wing, or aerostat) to harvest wind power from significantly increased altitudes (typically up to 600m). At those altitudes, winds are stronger and more consistent than ground-level winds.

The vast energy resource from high-altitude winds has attracted the attention of numerous research and commercial ventures over the past two decades (Altaeros Energies Website, 2017; Ampyx Power Website, 2017; Kitenenergy Website, 2017; Makani Power, 2017; SkySails GmbH Website, 2017; Windlift, 2017). Besides being able to operate at much higher altitudes than traditional turbines, AWE systems also provide additional control degrees of freedom that allow the systems to *adjust* their operating altitudes and intentionally induce *crosswind motions* to enhance power output. To-date, the majority of research in the area of AWE system control has focused on the latter problem, i.e., crosswind

motion control (Canale, Fagiano, & Milanese, 2010; Fagiano, Milanese, & Piga, 2012; Isaacs, Hoagg, Hussein, & Olinger, 2011; Vermillion, Grunnagle, & Kolmanovsky, 2012; Williams, Lansdorp, & Ockels, 2007). Comparatively fewer studies have focused on the impact of adjusting altitude (Bafandeh & Vermillion, 2016; Vermillion, 2013). However, the first two papers make the assumption of a monotonic wind profile that conforms to a power law model, an assumption that greatly simplifies the altitude optimization problem but is seldom satisfied in the *instantaneous* wind shear (wind speed vs. altitude) profile (as we shall see in this paper). The more recent publication, Bafandeh and Vermillion (2016), does take into account the possibility of non-monotonic wind shear profiles but uses an extremum seeking (ES) formulation that only achieves convergence to *local*, rather than global, optima.

Only recently has any effort been undertaken to take into account the *stochastic* nature of the wind shear profile in the context of a *global* altitude optimization. An initial attempt to address this problem using model predictive control (MPC) is detailed in Bin-Karim, Bafandeh, and Vermillion (2016) and its expanded journal version (Bin-Karim, Bafandeh, Baheri, & Vermillion, submitted for publication). In the

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Fig. 1. Altaeros Buoyant Airborne Turbine (BAT), Image Credit: Altaeros Energies Website (2017).

context of the altitude optimization problem, MPC attempts to balance exploration with exploitation in optimizing the altitude setpoint. However, this MPC formulation requires an offline characterization of the statistical properties (conditional mean and conditional variance) of the wind shear (wind speed vs. altitude) profile, thereby necessitating the collection of a very substantial amount of data offline in order to kick-start the optimization.

For the altitude optimization problem at hand, it is desirable to employ a control system that can learn the statistical properties of the wind shear profile online, thereby alleviating the need for offline data collection prior to running the control system. One of the most well-studied problems in the machine learning community is the design of optimization algorithms for a real-world applications using scarce data. In existing literature, this problem has been studied in the context of sequential decision-making problems where the goal is to learn the behavior of an objective function (called exploration) while simultaneously trying to maximize or minimize the objective function (called exploitation). As an efficient and systematic approach to balance exploration and exploitation, Bayesian Optimization has been applied to various real-world problems (Abdelrahman, Berkenkamp, Poland, & Krause, 2016; Baheri, Deese, & Vermillion, 2017; Baheri, Ramaprabhu, & Vermillion, 2017; Calandra, Gopalan, Seyfarth, Peters, & Deisenroth, 2014; Garnett, Osborne, & Roberts, 2010). In general, Bayesian Optimization aims to blend exploration and exploitation in such a way that it finds the global optimum of an unknown, expensive-to-evaluate, and black-box function within only a few evaluations. One popular approach is to model the unknown function as a Gaussian Process (GP), where Bayesian Optimization puts prior belief the overall structure of that objective function. At every step of Bayesian Optimization, the next operating point is selected to maximize some *acquisition function*, which characterizes (i) how much will be learned by visiting a candidate point (exploration) and (ii) what the likely performance level will be at that next candidate point (exploitation) (Brochu, Cora, & De Freitas, 2010; Snoek, Larochelle, & Adams, 2012).

In the case of the altitude optimization problem at hand, the decision variable is the operating altitude at the next time step, denoted by z_{k+1} , where k is the current time step. The objective is net power output, taking under consideration the power required to control the system at

a given altitude and the power required to adjust the altitude. In this work, we focus specifically on the Altaeros Energies Buoyant Airborne Turbine (BAT), pictured in Fig. 1. The BAT maximizes power output by *hunting* for the optimal altitude.

This paper represents a greatly expanded version of our original conference publication, Baheri and Vermillion (2017), which provides an initial overview of the Bayesian Optimization approach and some initial results for the AWE application. The present paper provides additional details with regard to the Bayesian Optimization formulation and lower-level control structure that were omitted from Baheri and Vermillion (2017) for the sake of brevity. Additionally, this paper expands the results of Baheri and Vermillion (2017) through the following contributions:

- We compare the performance of Context-Dependent Bayesian Optimization (CDBO) with the three different acquisition functions introduced in Section 3.1.2.
- We present an analysis of the proposed algorithm under varying rate limits (which dictates the maximum allowable altitude change from one time step to the next), using a full week of real wind shear data.

The remainder of this paper is structured as follows: In Section 2, we summarize the wind shear profile data and formalize the objective function for the altitude optimization problem. Section 3 introduces the fundamentals of Gaussian Process (GP) modeling, Bayesian Optimization, and the proposed algorithm. Section 4 provides detailed results, indicating significant improvement in net energy generation over baseline approaches.

2. Wind shear profile, altitude adjustment, and energy generation models

2.1. Wind shear profile

To maximize generated energy, it is of interest to operate the AWE system at altitudes where the wind velocity, V_{wind} , is as close as possible to the rated wind speed, V_{rated} . Consistent with intuition, if $V_{\text{wind}} < V_{\text{rated}}$, the net power production will diminish because the turbine is producing less power. Less obviously, the net power production will also diminish whenever $V_{\text{wind}} > V_{\text{rated}}$ because more power will be expended controlling the system. While it is possible to install wind profiling equipment to continuously monitor V_{wind} as a function of altitude (z), such profiling equipment is very costly. Table 1 shows an estimated cost comparison between different wind profiling technologies, including fixed LiDAR and SODAR units, as well as a weather balloon that continually is raised and lowered through the domain of allowable altitudes. These costs account for capital expenses, along with operational expenses that include the cost of the energy required to operate the system. The cost estimate for the weather balloon system includes not only the balloon, but also the on-board instrumentation and winch system that continually actuates the weather balloon. Given these costs, it is of economic interest to use the AWE system itself to periodically explore the wind shear environment, taking into account the spatio-temporal variability of the wind in deciding which altitude to explore next. In the case of the Altaeros BAT (used as a case study in this work), the wind speed is measured through the use of an ultrasonic anemometer, as discussed in Vermillion, Grunnagle, Lim, and Kolmanovsky (2014). Anemometer measurements (which provide apparent wind speed) are used in conjunction with GPS data to compute the true wind speed. This alleviates the need for expensive equipment but only provides a wind speed measurement at the AWE system's operating altitude.

To conduct our study, we used data obtained from a Doppler radar-based wind profiler in Cape Henlopen State Park (located in Lewes, Delaware). The data includes wind speed at 30 min intervals, at 48 altitudes up to 3000 m, over the course of one year (Archer, 2014). Due to technical constraints for AWE system flight altitude, we used

Table 1
20-year cost comparison for different technologies of measuring wind shear profile.

Technology	Capital cost	Annual operational cost	Total cost
SODAR	990,000	36,000	1,710,000
LiDAR	190,000	32,000	830,000
Weather Balloon + Winch System	50,000	1,000	70,000

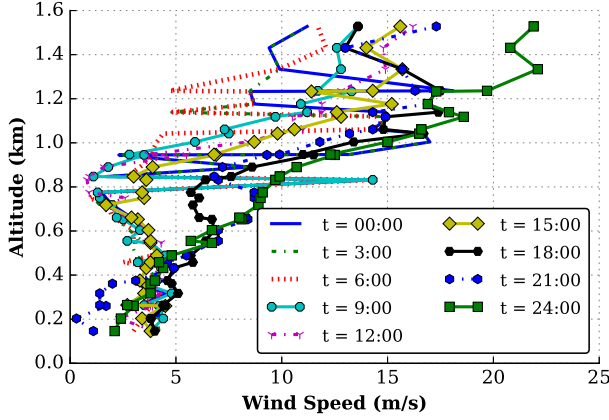


Fig. 2. Sample wind shear profile based on actual data for one day from Cape Henlopen State Park, Lewes, Delaware, Date: March 1, 2014, Archer (2014). The propagation of wind shear profile at 3 h increments has been shown. One can conclude that (i) the optimal altitude varies over the course of the day and (ii) the wind shear profile does not fit a neat, monotonic relationship.

the portion of data involving altitudes up to roughly 1600 m (one mile) above ground level for the study at hand.

The progression of the wind shear profile over the course of one day, at 3 h intervals, is shown in Fig. 2. One can infer from this figure that the wind profile at any given time does not follow a power law or logarithmic law wind profile (or even a monotonic trend in many instances). Thus, in order to optimize net energy capture, it is imperative to employ a control scheme that is capable of coping with the uncertain nature of the wind shear profile.

2.2. Altitude adjustment model

In any AWE system, a low-level flight control system must exist that regulates key flight variables, such as altitude, pitch angle, and roll angle, to prescribed setpoints. The block diagram of the low-level control structure used for the AWE system considered in this work (the Altaeros BAT) is shown in Fig. 3 and is described in detail in Vermillion et al. (2014). Among other things, Vermillion et al. (2014) demonstrates that an underlying flight control system can successfully control the BAT to prescribed altitude setpoints, denoted by z_{sp} . Because the efficacy of this lower-level controller has already been demonstrated in the literature and because of the relatively long time steps involved with altitude optimization, we will work in this paper with a very simple altitude adjustment model that assumes that *the altitude at the next step is equal to the altitude setpoint from the current time step*, i.e.:

$$z_{t+1} = z_t^{sp}. \quad (1)$$

Here, z_t^{sp} denotes the altitude setpoint at step t . Given the altitude tracking performance demonstrated in Vermillion et al. (2014) and the long time step of 30 min that we will work with for the adjustment of z^{sp} in this paper, this very simple model represents a reasonable approximation. In fact, a lower-level control has been designed in Vermillion et al. (2014) for altitude tracking, and this lower-level controller operates on a much faster time step. In this paper, we evaluate the impact of different rate limitations (i.e., limitations on $z_{t+1}^{sp} - z_t^{sp}$) that ensure that any prescribed altitude adjustment can be made quickly (relative to the

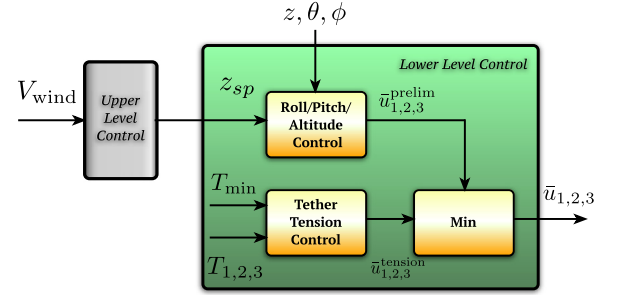


Fig. 3. Lower-level controller Altaeros BAT, as detailed in Vermillion et al. (2012). $\bar{u}$ represents the vector of control inputs to the winches, whereas T_i denotes the tension in each tether.

total 30 min time step for altitude optimization) and the vast majority of time is spent generating energy at a prescribed altitude rather than adjusting altitude. The impact of different rate limitations on control results is studied in detail in Section 4.

2.3. Energy generation model

In this section, we formalize the objective function of altitude optimization. An appropriate objective function for altitude optimization should account for the energy generated by the turbine and the energy lost by adjusting and maintaining altitude (even when maintaining altitude, control energy must be expended to make small adjustments to the tethers in order to reject typical levels of turbulence). In order to account for each of these objectives, the instantaneous net power generation is expressed as Bafandeh and Vermillion (2016):

$$P_{total} = c_1 \min(V_{wind}, V_{rated})^3 - c_2 V_{wind}^2 + P_z(V_{wind}, \dot{z}). \quad (2)$$

Generally, the power produced by a wind turbine can be modeled as:

$$P_{tur} = \frac{1}{2} \rho A V_{wind}^3 C_P, \quad (3)$$

where ρ is the air density, A is the rotor area, and C_P is the power coefficient. Accordingly, c_1 can be defined as:

$$c_1 = \frac{P_{tur}}{V_{wind}^3} = \frac{1}{2} \rho A C_P. \quad (4)$$

To calculate c_2 , it is important to consider that maintaining altitude requires the tethers to apply a total force that overcomes the lifting force on the BAT. The application of this force alone does not require energy. However, in the presence of turbulence (or any wind environment that is less than perfect), small adjustments are necessary to maintain altitude. The power required to make these adjustments is proportional to the tension in the tethers multiplied by the tether reel-in speed. The lifting force on the aerostat body is given by:

$$L = \frac{1}{2} \rho A_{ref} V_{wind}^2 C_L + F_b, \quad (5)$$

where L is the total lift force, A_{ref} is the reference area for the BAT, C_L is the lumped coefficient of lift, and F_b is the buoyant force. Under high wind conditions (under which the tether tensions are significant), F_b is much smaller than the aerodynamic lifting force, and the lifting force can be approximated by:

$$L = \frac{1}{2} \rho A_{ref} V_{wind}^2 C_L. \quad (6)$$

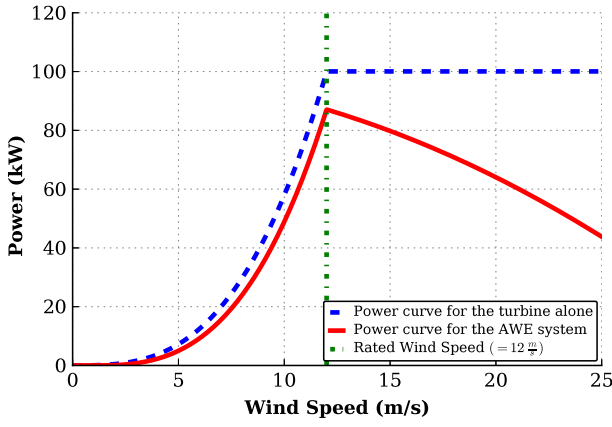


Fig. 4. Comparison of power curve for the turbine alone and the full AWE system. The power production decreases for the wind velocity beyond V_{rated} due to increased energy required to control system to an altitude set-point in the presence of typical wind disturbances.

Since the power expenditure for maintaining altitude is proportional to this lifting force multiplied by the tether speed, that power expenditure, P_{maint} , can be expressed as:

$$P_{\text{maint}} = \frac{1}{2} \rho A_{\text{ref}} V_{\text{wind}}^2 V_{\text{ave}} C_L, \quad (7)$$

where V_{ave} is the average tether release speed over typical operation. Accordingly, c_2 can be defined as:

$$c_2 = \frac{P_{\text{maint}}}{V_{\text{ave}}^2} = \frac{1}{2} \rho A_{\text{ref}} V_{\text{ave}} C_L. \quad (8)$$

The power required to adjust altitude, P_z , is given as follows:

$$P_z(V_{\text{wind}}, \dot{z}) = \begin{cases} \frac{c_3}{\eta_m} \dot{z} V_{\text{wind}}^2 & \dot{z} \leq 0, \\ c_3 \eta_g \dot{z} V_{\text{wind}}^2 & \text{otherwise.} \end{cases} \quad (9)$$

where η_m and η_g represent the efficiency of the motors that drive the winches and efficiency of the winches, respectively. The parameter c_3 is a constant that depends on the aerodynamics of the lifting body and its trim pitch angle. Because any upward motion of the AWE system must at some later time be compensated by downward movement (the AWE system must land eventually), Bafandeh and Vermillion (2016) presents a simplified version of Eq. (2), which will be used in this work and is given by:

$$P = c_1 \min(V_{\text{wind}}(z), V_{\text{rated}})^3 - c_2 V_{\text{wind}}^2 - \bar{c}_3 V_{\text{wind}}^2 |\dot{z}|. \quad (10)$$

Here, $\bar{c}_3$ is a lumped parameter that is dependent on overall winch efficiency and aerodynamics of the lifting body. In our discrete-time altitude optimization algorithm, we approximate $|\dot{z}_k|$ by:

$$|\dot{z}_k| = \frac{|z_k - z_{k-1}|}{\Delta t} \quad (11)$$

where Δt denotes the time step.

Ultimately, Eq. (10) serves as our cost function to be maximized. The AWE system power curve is shown in Fig. 4. This figure illustrates the power curve for a 100 kW variable-speed turbine, which is in line with the BAT design. One can conclude from this figure that the power generated by the full AWE system is always less than the power generated by the turbine alone, due to the energy that must be used to control the system's altitude to a constant set-point, under typical turbulence levels. Moreover, unlike the conventional wind energy systems, the power generated decreases for wind speeds that exceed V_{rated} . This is the result of higher aerodynamic loads, which in turn requires more energy to control the system to an altitude set-point in the presence of typical turbulence levels. It should be noted that Fig. 4 is specific to

scenarios where the AWE system is stationary (altitude is not changing); additional energy losses exist when the altitude changes from one time step to the next.

3. Methodology

The final goal of altitude optimization with Bayesian Optimization is to maximize the energy generated by the BAT. Due to partially observable nature of wind shear profile, the optimal altitude in not known *a priori*. Hence, the optimization strategy must appropriately balance a trade-off between learning more about the objective function (exploration) and selecting an altitude that maximizes expected net power generation based on the collected data (exploitation).

3.1. Bayesian optimization

Suppose that we want to maximize an unknown, black-box, and expensive-to-evaluate objective function. Due to the cost associated with evaluating this function, it is crucial to select the location of each new evaluation deliberately. For example, in the case of the altitude optimization problem at hand, the task of moving the AWE system to a new altitude is very expensive in terms of time, energy, and cost. In such cases, Bayesian Optimization algorithm serves as an attractive technique for finding the *global optimum* of an unknown function within *few evaluations*. The major assumption here is that the objective function evaluation is costly, whereas computational resources are inexpensive. This exactly matches the altitude optimization under consideration, where each evaluation of the objective function corresponds to an expensive experiment that requires physically relocating the AWE system to a new altitude.

Broadly speaking, Bayesian Optimization involves two steps. First, at each iteration, we update a model that characterizes our “best guess” at the objective function vs. design parameter (i.e., altitude), along with a *quantification* of the uncertainty in that guess. This phase of Bayesian Optimization is referred to as the *learning phase*. Second, we choose an *acquisition function*, which *guides* the optimization by determining the next point (i.e., next altitude) to evaluate. The selection of the next altitude in an effort to maximize the acquisition function is referred to as the *optimization phase*.

3.1.1. Learning phase: using Gaussian Processes (GPs)

In this section, we introduce the basic properties of GPs. GPs are used to model the objective function to be maximized.

As an attractive choice for non-parametric regression in machine learning, the ultimate goal of GP models is to find an approximation of a complex map from an input vector to the objective function value. In general, GP models are able to model complex phenomena such as nonlinear effects and interaction between covariates in a systematic fashion. GP models assume that the objective function values associated with different inputs (i.e., altitude) have a joint Gaussian distribution (Rasmussen & Williams, 2006).

In general, a GP is fully specified by its mean function, $\mu(z)$, and covariance function, $k(z, z')$:

$$P(z) \sim \mathcal{GP}(\mu(z), k(z, z')). \quad (12)$$

In the context of the altitude optimization problem, the GP framework is used to predict the generated power, $P(z)$, at candidate altitude z_c , based on a set of t past observations, $D_{1:t} = \{z_{1:t}, P(z_{1:t})\}$. In general, having $\mu(z) = 0$ significantly simplifies the training and prediction phase of a GP model (Rasmussen & Williams, 2006). To achieve this in the AWE application, we normalize the outputs of GP model using *z-score* normalization. Thus, the normalized power at step $t + 1$, denoted by $\bar{P}_{t+1}(z)$, is expressed as:

$$\bar{P}_{t+1}(z) = \frac{P_{t+1}(z) - \mu^{1:t}}{\sigma^{1:t}} \quad (13)$$

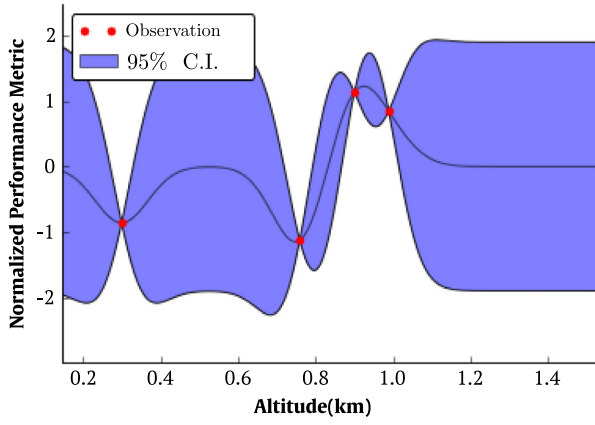


Fig. 5. Gaussian Processes (GP) models can both make predictions and maintain a measure of uncertainty over those predictions. The 95% confidence of the model prediction is represented by blue area. Once an observation is made at a particular point, the uncertainty drops to zero. Recognizing that the wind shear profile is time-varying, Section 3.2 discusses how context-dependent Bayesian Optimization can be used to account for the fact that uncertainty increases in relation to the amount of time that has passed since an observation has been made (i.e., under context-dependent Bayesian Optimization, uncertainty at a location does not remain zero for all time just because that location was visited in the past). (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

where $\mu^{1:t}$ and $\sigma^{1:t}$ denote the mean and standard deviation of the population up to step t .

The normalized power at candidate altitude z_c , represented by $\bar{P}(z_c)$ in Eq. (14), for an unobserved altitude, z_c , is modeled to follow a multivariate Gaussian distribution (Rasmussen & Williams, 2006):

$$\begin{bmatrix} y_{1:t} \\ \bar{P}(z_c) \end{bmatrix} \sim \mathcal{N}\left(0, \begin{bmatrix} K_t + \sigma_\epsilon^2 I_t & k_t^T \\ k_t & k(z_c, z_c) \end{bmatrix}\right), \quad (14)$$

where $y_t = \{\bar{P}(z_1), \dots, \bar{P}(z_t)\}$ is the vector of observed function values. The vector $k_t(z) = [k(x^*, x_1), \dots, k(x^*, x_t)]$ encodes the covariances between the candidate altitude, z_c , and the past data points, $z^{1:t}$. The past-data covariance matrix, with entries $[K_t]_{(i,j)} = k(z_i, z_j)$ for $i, j \in \{1, \dots, t\}$, characterizes the covariances between pairs of past data points (Fig. 5). The identity matrix is represented by I_t , and σ_ϵ represents the noise variance (Rasmussen & Williams, 2006).

Individual elements of the covariance matrix, namely $k(z_i, z_j)$, encode the correlation between two different altitudes. In order to characterize this correlation in a simple, closed-form manner, a *covariance kernel* is used. This covariance kernel provides a relatively simple parametric structure for the values of $k(z_i, z_j)$. In this work, we represent the elements of the covariance matrix through a very commonly used kernel function known as the Squared Exponential (SE) covariance kernel. For two altitudes, z_i and z_j , the SE kernel is parameterized as:

$$k(z_i, z_j) = \sigma_0^2 \exp\left(-\frac{1}{2}(z_i - z_j)^T \Lambda^{-2}(z_i - z_j)\right) + \sigma_\epsilon \delta_{ij}, \quad (15)$$

where δ_{ij} represents the Kronecker delta operator:

$$\delta_{ij} = \begin{cases} 1, & i = j, \\ 0, & i \neq j. \end{cases} \quad (16)$$

We choose the SE kernel based on observation of key characteristics of the problem at hand. Qualitatively, the SE kernel structure accounts for the fact that two measurements taken at close altitudes are more likely to be correlated than measurements taken at distant altitudes. Furthermore, the SE kernel accounts for the fact that the value of data decays exponentially with variations in time and altitude, which has been generally seen to be reflected in actual wind shear data. In fact, the assessment performed in Bin-Karim et al. (2016) shows that the level of certainty in the wind speed decays exponentially with respect to both time and position.

The quantitative correlation between different data points is characterized by *hyper-parameters*, which are denoted by $\theta = \{\sigma_0, \Lambda, \sigma_\epsilon\}$. Kernel hyper-parameters are identified by maximizing the marginal log-likelihood of the existing observed data, D (Rasmussen & Williams, 2006):

$$\theta^* = \arg \max_{\theta} \log p(y_t | z^{1:t}, \theta), \quad (17)$$

where

$$\log p(y_t | z^{1:t}, \theta) = \left(-\frac{1}{2} y_t^T K_t^{-1} y_t - \frac{1}{2} \log |K_t| - \frac{t}{2} \log 2\pi\right). \quad (18)$$

Once the hyper-parameters are optimized, the predictive mean and variance at z^* , conditioned on these past observations, are expressed as:

$$\mu_t(z^* | D) = k_t(z^*) \left(K_t + I_t \sigma_\epsilon^2\right)^{-1} y_t^T, \quad (19)$$

$$\sigma_t^2(z^* | D) = k(z^*, z^*) - k_t(z^*) \left(K_t + I_t \sigma_\epsilon^2\right)^{-1} k_t^T(z^*). \quad (20)$$

3.1.2. Acquisition function

Bayesian Optimization bases its choice of next operating point on the maximization of an *acquisition function*, which guides the optimization by determining the next point to evaluate. Specifically, the acquisition function uses the predictive mean and variance (Eqs. (19)–(20)) to combine exploration (high-variance regions) and exploitation (high-mean regions) in selecting the next operating point. We denote the acquisition function by $\alpha(\cdot)$ in this work.

To-date, several acquisition function structures have been proposed in the literature, including probability of improvement, expected improvement, upper confidence bound, information gain, and entropy-based improvements. All acquisition functions have a common theme: they incorporate both the mean, μ_t , and the variance, σ_t^2 , of the GP model prediction. As a result, each technique leads to a trade-off between exploration and exploitation. In this work, we will consider three acquisition function structures for the altitude optimization problem.

• Maximum Probability of Improvement (MPI)

The first and most intuitive acquisition function proposed for Bayesian Optimization is Maximum Probability of Improvement (MPI) (Kushner, 1964). This acquisition selects the next operating point as the one that is most likely to improve upon the best value of objective function so far. Mathematically, the acquisition function takes the following form:

$$\alpha_{PI}(z) \triangleq p\left(P(z) \geq P(z)^{\max}\right), \quad (21)$$

where $P(z)^{\max}$ represents the best value of objective function so far. Thus, the MPI acquisition function can be expressed as:

$$\alpha_{PI}(z) = \phi\left(\frac{\mu_t(z) - P(z)^{\max}}{\sigma_t(z)}\right), \quad (22)$$

where $\phi(\cdot)$ denotes the normal cumulative distribution function (CDF). The point with the highest probability of improvement is selected as the best next point to evaluate.

• Expected Improvement (EI)

In many practical applications, the MPI acquisition function results in local optima, since it does not take into account the size of the improvement in its formulation. As an alternative to the MPI acquisition function, the Expected Improvement (EI) acquisition function (Brochu et al., 2010; Zilinskas, Mockus & Tiesis, 1978) considers the *amount* of anticipated improvement when choosing the next operating point.

Algorithm 1 Bayesian Optimization (BO)

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1: procedure ALTITUDE OPTIMIZATION WITH BO
2:    $D \leftarrow \text{Initialize: } \{z_{1:2}, P(z_{1:2})\}$ 
3:   for each iteration do
4:     Train a GP model from  $D$ 
5:     Compute mean and variance of GP (Eqs. (19)–(20))
6:     Compute acquisition function (see Sec. 3.1.2)
7:     Find  $z^*$  that optimizes acquisition function
8:     Append  $\{z^*, P(z^*)\}$  to  $D$ 

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To construct the EI acquisition function, the improvement function is defined by:

$$I(z) \triangleq \max \{0, P(z) - P(z)^{\max}\}. \quad (23)$$

If the prediction is higher than the best value of the objective function so far, then the improvement is positive. Otherwise, it is set to zero. The inability for the acquisition function to assume negative values reflects the fact that if the solution *worsens* from one iteration to the next, then it is possible to simply revert to the previous best solution. The EI acquisition function is given by:

$$\alpha_{EI}(z) = \mathbb{E}(I(z)) = \mathbb{E}[\max \{0, P(z) - P(z)^{\max}\}], \quad (24)$$

where the later equation can be expressed as:

$$\alpha_{EI}(z) = \int_{P(z)^{\max}}^{\infty} \underbrace{\max \{0, P(z) - P(z)^{\max}\}}_{\text{value of improvement}} \cdot \underbrace{\phi\left(\frac{\mu_t(z) - P(z)^{\max}}{\sigma_t(z)}\right)}_{\text{probability of improvement}} dP. \quad (25)$$

The first term in the integrand of Eq. (25) is the amount of improvement, and the second term represents the probability of that improvement. Calculating the integral of Eq. (25) using integration by parts leads to EI acquisition function (Jones, Schonlau, & Welch, 1998) that is given by:

$$\alpha_{EI}(z) = \begin{cases} \left(\mu_t(z) - P(z)^{\max}\right)\Phi(Z) + \sigma_t(z)\phi(Z), & \sigma_t(z) > 0 \\ 0, & \sigma_t(z) = 0 \end{cases} \quad (26)$$

where:

$$Z = \frac{\mu_t(z) - P(z)^{\max}}{\sigma_t(z)}, \quad (27)$$

and $\Phi(\cdot)$ denotes the probability density function (PDF) for the normal distribution. The next operating point is chosen as the one that maximizes α_{EI} .

- **Upper Confidence Bound (UCB)**

The final candidate acquisition function considered in this work is known as the *upper confidence bound* (UCB) acquisition function (Cox & John, 1997), which incorporates the mean and variance *explicitly* at a particular location:

$$\alpha_{UCB}(z) \triangleq \mu_t(z) + \sqrt{\beta_t} \sigma_t(z), \quad (28)$$

β_t plays a crucial role in Eq. (28), as it determines the trade-off between exploration and exploitation. A special case of UCB is GP-UCB, where β_t at each time step is automatically set according to Srinivas, Krause, Kakade, and Seeger (2009):

$$\beta_t = 2 \log\left(\frac{|D|t^2\pi^2}{6\delta}\right). \quad (29)$$

Here, t is the number of past evaluations of the objective function, $\delta \in (0, 1)$ is a calibration parameter, and D is the number of decision variables.

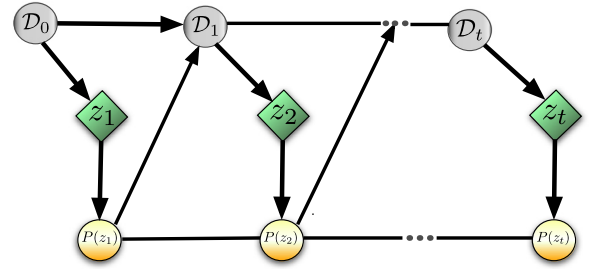


Fig. 6. A graphical network (inspired from González, Osborne, and Lawrence, 2015) illustrating the Bayesian Optimization concept. Given some initial data, D_0 , Bayesian Optimization identifies the next best operating altitude, z_1 , to evaluate. The objective function is then evaluated at this altitude, $P(z_1)$. Next, this pair is augmented to the previous data, the model is updated, and the process is repeated.

Algorithm 2 Context-Dependent Bayesian Optimization (CDBO)

```

1: procedure ALTITUDE OPTIMIZATION WITH CDBO
2:    $D \leftarrow \text{Initialize: } \{z_{1:2}, t_{1:2}, P(z_{1:2})\}$ 
3:   for each time step do
4:     Set the context into the current time
5:     Construct a composite kernel (Eq. (30))
6:     Train a GP model from  $D$ 
7:     Compute predictive mean and variance of GP
8:     Compute acquisition function (see Sec. 3.1.2)
9:     Find  $z^*$  that optimizes acquisition function
10:    Append  $\{(z^*, t_{\text{current time}}), P(z^*)\}$  to  $D$ 

```

We summarize the conventional Bayesian Optimization in Algorithm 1. The algorithm is initialized by two previously evaluated altitudes and corresponding function values (line 2). Then, at each step, a GP model is trained (line 4) to compute the predictive mean and variance (line 5). These statistical quantities are used to construct the acquisition function (line 6). Next, the point that maximizes the acquisition function is selected as the next altitude (line 7). Finally, data from the chosen operating point is added to the historical data (line 8), and the process is restated. Fig. 6 illustrates the process outlined here.

Fig. 12 illustrates the evolution of Bayesian Optimization process for a course of one day, based on Algorithm 1, for the altitude optimization problem. The horizontal axis represents the altitude space. The vertical axis denotes the performance, normalized by z -score (Eq. (13)). The red dots depict the suggested operating altitude, and the blue area represents the 95% confidence bound of the GP model. From the GP model, the acquisition function (red curve) is computed. The point that maximizes the acquisition function is selected as the next altitude (red dots). The evolution process has been shown for every one hour for a course of one day.

3.2. Context-dependent Bayesian optimization

The time-varying nature of our objective function compels us to modify the conventional Bayesian Optimization method (Algorithm 1). In fact, Algorithm 1 assumes that the objective function does not change during the optimization procedure; specifically, Algorithm 1 assumes that once an altitude has been visited, the uncertainty surrounding the wind speed at that altitude is equal to zero. Due to the dynamic nature of the objective function, however, this is not the case in reality. One way to cope with the dynamic nature of objective function is through contextual GP modeling (Krause & Ong, 2011). Contextual GP modeling allows for the possibility that the objective function depends on both the operating point (altitude in our application) and *other environmental variables* (termed the “context”). To deal with time-varying nature of our objective function, we consider *time* as the context.

Employing contextual GP is very straightforward. In Section 3.1.1, the kernel of the GP is defined only in terms of altitude. With contextual

Table 2
Cost function parameters.

Parameter	Description	Value
c_1	Coefficient of power production related to the power curve	0.0579
c_2	Coefficient of power production required to maintain altitude	0.09
c_3	Coefficient of power production required to change altitude	0.15

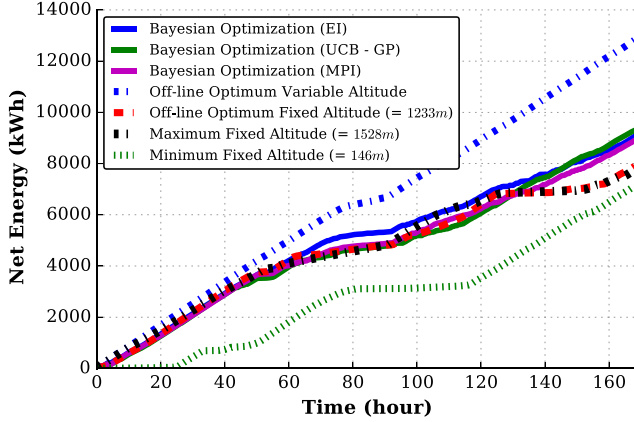


Fig. 7. Comparison of net energy production over the course of one week. Bayesian Optimization (with different acquisition functions) outperforms off-line optimized fixed altitude, minimum fixed altitude, and maximum fixed altitude.

GP, the kernel is also allowed to vary as a function of the context (time). For this purpose, we choose an addition of two kernels, one over altitude, k_z , and one over time, k_t

$$k((z_i, t_i), (z_j, t_j)) = k_z(z_i, z_j) + k_t(t_i, t_j), \quad (30)$$

where:

$$k_z(z_i, z_j) = \sigma_{0z}^2 \exp\left(-\frac{1}{2}(z_i - z_j)^T \Lambda_z^{-2}(z_i - z_j)\right) + \sigma_{\epsilon z} \delta_{ij}, \quad (31)$$

and

$$k_t(t_i, t_j) = \sigma_{0t}^2 \exp\left(-\frac{1}{2}(t_i - t_j)^T \Lambda_t^{-2}(t_i - t_j)\right) + \sigma_{\epsilon t} \delta_{ij}. \quad (32)$$

In Eqs. (31) and (32), $\theta_z = \{\sigma_{0z}, \Lambda_z, \sigma_{\epsilon z}\}$ and $\theta_t = \{\sigma_{0t}, \Lambda_t, \sigma_{\epsilon t}\}$ denote the hyper-parameters of k_z and k_t , respectively. These hyper-parameters describe how quickly the wind shear profile changes and how similar one altitude is to the next at a given time. The collection of hyper-parameters are automatically *tuned* (i.e., optimized) at each iteration of our algorithm, based on the data up to that time, using a maximum likelihood estimation criteria (see Eq. (18)).

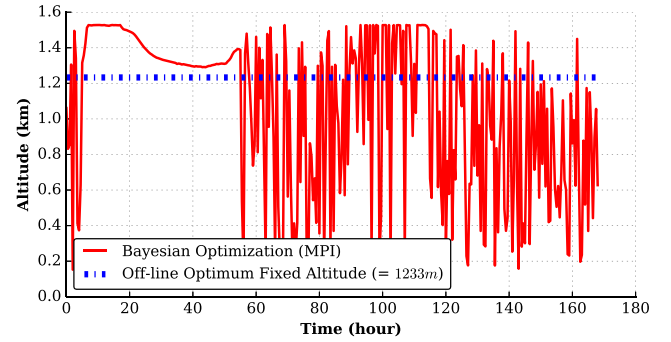
Context-Dependent Bayesian Optimization is summarized in Algorithm 2. First, at the very first two time steps, the procedure is initialized by two previously evaluated altitudes and the corresponding function values (line 2). Then, at each time step, the context (time) is fixed at the current instance (line 4). Next, the composite kernel is constructed based upon Eq. (30) (line 5). Similar to Algorithm 1, the GP model, along with its predictive mean and variance, is computed (line 6–7). Then, the acquisition is computed by one of the methods outlined in Section 3.1.2, and optimization is done only over the altitude space (line 8–9). Finally, the selected altitude and the current time are augmented to the data (line 10), and the process is restarted.

4. Results

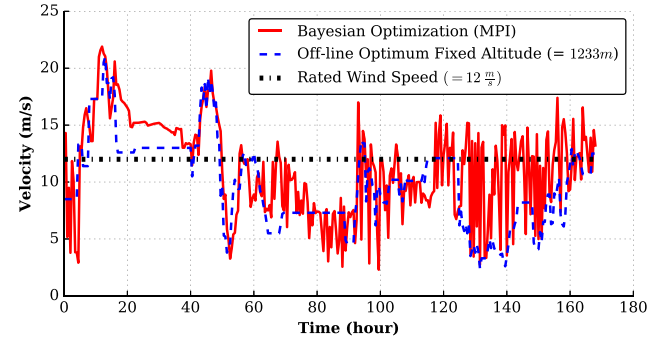
We evaluate the proposed altitude optimization algorithm by simulating the model in Section 2.3 based on available data (Archer, 2014). We implemented our algorithm with the Python language on a PC-based workstation (2.4 GHz intel Core i5, 4GB RAM). For this work,

Table 3
Design parameters for Bayesian Optimization algorithm.

Parameter	Description	Value
P_{rated}	Rated power of turbine	100 kW
V_{rated}	Rated wind velocity	12 $\frac{m}{s}$
z_{min}	Minimum allowable altitude	146 m
z_{max}	Maximum allowable altitude	1528 m
Δt	Time step	30 min



(a) Comparison of operating altitude.



(b) Comparison of wind speeds at operating altitude.

Fig. 8. Comparison of operating altitude and wind velocity over the course of one week using Maximum Probability of Improvement (MPI) acquisition function.

simulations were run in a non-real-time environment (the simulation of one week of data takes less than 5 min).

Tables 2 and 3 summarize the objective function parameters and design parameters, respectively. We compare the system performance under four different control scenarios:

- **Scenario 1: Off-line Optimum Fixed Altitude:** The first scenario involves flying the AWE at a *optimum fixed altitude*. We identify this optimal fixed altitude by off-line calculation of energy generation at different altitudes, using given wind data. Thus, the optimization takes the following form:

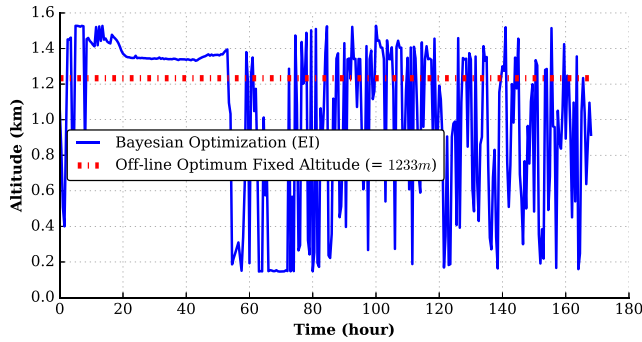
$$\underset{z=\text{constant}}{\text{maximize}} \quad \sum_{t=0}^{\tau} P_t(z)$$

where τ is the final time step. It is important to note that this calculation requires knowledge of the future. Hence, this scenario serves an upper bound on the potential of any fixed altitude algorithm but is not implementable in practice. The off-line

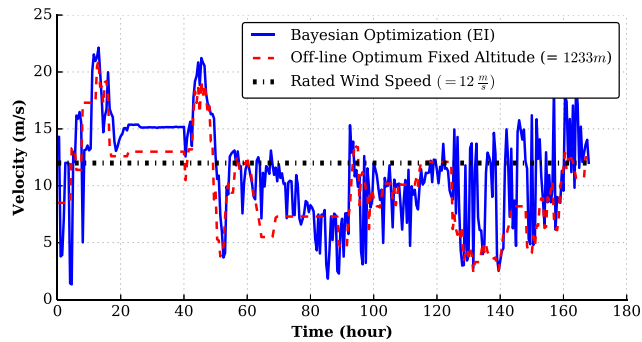
Table 4

Comparison of net energy production for 10 randomly selected weeks.

	Bayesian Optimization [kWh]	Off-line optimum fixed altitude [kWh]	Maximum fixed altitude [kWh]	Minimum fixed altitude [kWh]
Week 1	9371	8990	8688	6200
Week 2	9068	7896	7752	7111
Week 3	8774	7818	7809	7811
Week 4	9441	9374	9214	6913
Week 5	8934	8920	8442	7414
Week 6	11 306	10 867	7752	6786
Week 7	8827	8514	8278	7349
Week 8	10 804	11 019	10 721	6888
Week 9	8706	8714	8693	7634
Week 10	9586	9872	9626	7060
Mean	9481.7	9198.4	8697.5	7116.6

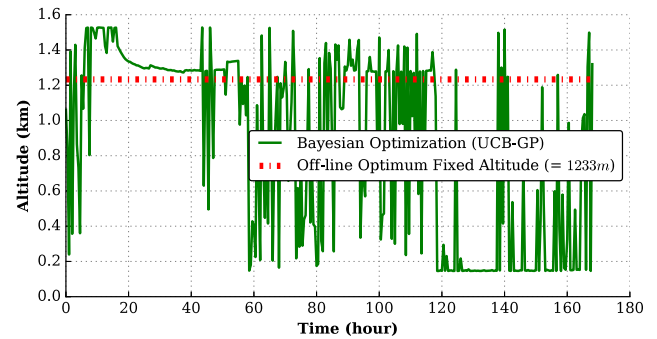


(a) Comparison of operating altitude.

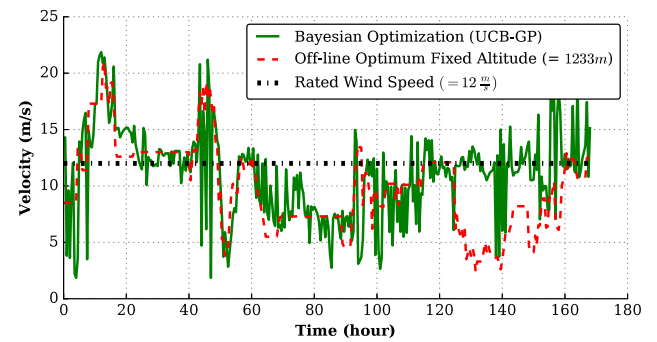


(b) Comparison of wind speeds at operating altitude.

Fig. 9. Comparison of operating altitude and wind velocity over the course of one week using Expected Improvement (EI) acquisition function.



(a) Comparison of operating altitude.



(b) Comparison of wind speeds at operating altitude.

Fig. 10. Comparison of operating altitude and wind velocity over the course of one week using Upper Confidence Bound (UCB-GP) acquisition function.

optimum fixed altitude was 1233 m for the time period studied in this work.

- **Scenario 2: Minimum Fixed Altitude:** The second scenario is representative of conventional towered wind turbines. For this scenario, we rely on data from the lowest available altitude 146 m, which is nearly equal to the hub length of the world's tallest towered wind turbines.
- **Scenario 3: Maximum Fixed Altitude:** The third scenario involves flying the AWE at maximum fixed altitude, 1528 m.
- **Scenario 4: Bayesian Optimization:** The fourth scenario employs Context-Dependent Bayesian Optimization (CDBO), as outlined in Algorithm 2.

The design parameters used in the study are reported in Table 3. Bayesian Optimization is conducted at each 30 min time interval to identify the best next operating altitude.

Table 4 compares the performance of Bayesian Optimization (i.e., Scenario 4) to baseline scenarios (i.e., Scenario 1–3) for 10 randomly selected weeks in our dataset. One can conclude from this table that our algorithm outperforms than other baseline scenarios

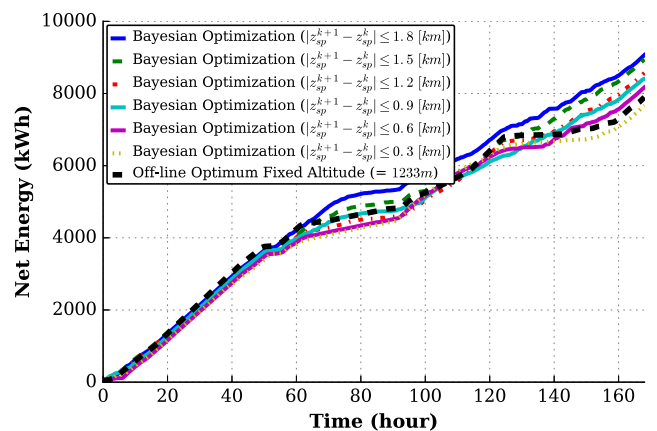


Fig. 11. Comparison of net energy production where the altitude adjustment under a variety of rate limitations.

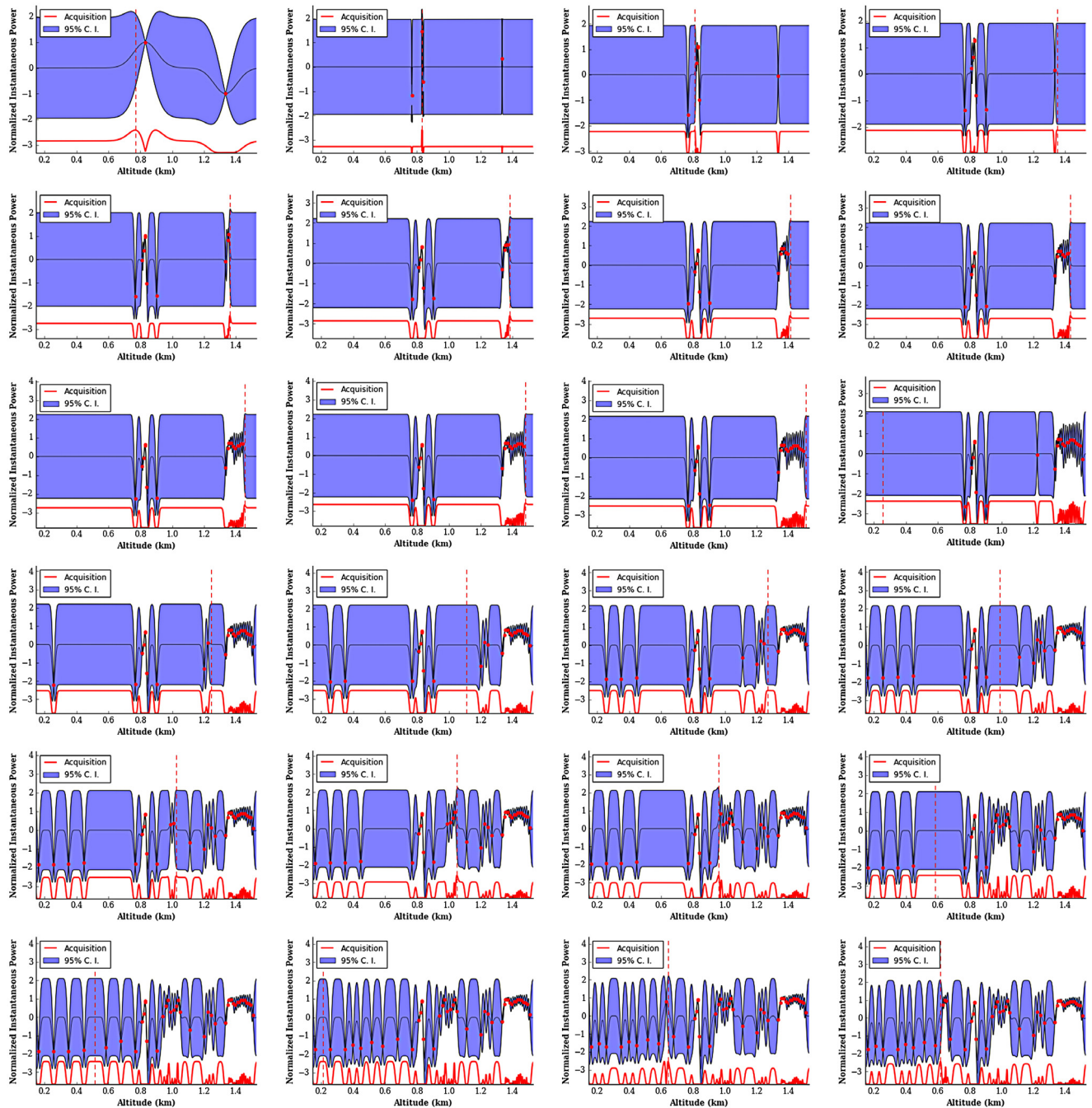


Fig. 12. Illustration of Bayesian Optimization (Algorithm 1) for a course of one day at one hour intervals (from upper left to lower right). The horizontal axis represents the altitude space. The vertical axis denotes the performance, normalized by z -test. The 95% confidence of the model prediction is represented by the blue area. The model is initialized with two previously evaluated altitudes and the corresponding function values. The location of the next altitude to be evaluated is determined by the acquisition function (vertical red line). At each time step, the model is updated using all the previously evaluated altitudes (red dots). (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

in most instances. Furthermore, Fig. 7 represents the performance of Bayesian Optimization over the course of one week for one of these selected weeks, using each of the three candidate acquisition functions outlined in Section 3.1.2. One can conclude from this figure that all Bayesian Optimization methods result in superior energy production when compared to baseline scenarios. In this figure, we have also included a new scenario termed the *off-line optimum variable altitude*. This scenario serves as an upper bound on the achievable energy output that cannot be exceeded under any control strategy. It should be noted that, similar to off-line optimum fixed altitude, this scenario cannot be implemented in real-time as it requires the knowledge of future. The

resulting operating altitude and corresponding wind velocity are shown in Figs. 8–10.

4.1. Exploration rate limitation

To account for different limitations on the tether release rate for the AWE system, Fig. 11 illustrates the net energy production of the AWE system under stricter rate limits than the one imposed in our initial result. One can infer from this figure that the most net energy occurs under the least severe rate limit, as expected. However, even strict rate limitations result in more energy production than constant-altitude control strategies.

5. Conclusions and future work

5.1. Conclusions

In this work, we presented a data-driven approach for finding the optimal altitude of AWE system that maximizes net power output. Furthermore, we extended the conventional Bayesian Optimization to capture time-dependent patterns of our objective function. Using real wind data, we showed that the use of context-dependent Bayesian Optimization improves the resulting total net energy generation, as compared to other control scenarios.

5.2. Future work

Future work will include:

- Considering non-myopic Bayesian Optimization, which can account for multiple future steps (instead of merely considering the next time step).
- Comparing the results of Bayesian Optimization with model-based approaches such as model predictive control, along with non-model based approaches such as extremum seeking.
- Implementing windowing to circumvent increasing the size of covariance matrix in the GP model with all incoming data.
- Considering utilization of dynamic structure of GP to model the dynamic nature of wind shear profile.
- Experimental validation of Bayesian Optimization on the Altaeros Buoyant Airborne Turbine (BAT).

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